ADCS Homework 2

John Zhang

I. Safe Mode

A. Inertia Perturbation

The ISS nominal principal moments of inertia are $I_{xx} = 1.280 \times 10^8\,\mathrm{kg\,m^2}$, $I_{yy} = 1.070 \times 10^8\,\mathrm{kg\,m^2}$, $I_{zz} = 2.010 \times 10^8\,\mathrm{kg\,m^2}$ [1]. To ensure the solar panel normal ($+Z$ body) is not aligned with a principal axis, the inertia is perturbed via eigendecomposition:

$$J = VDV^T, \quad \tilde{D} = D(I + \mathrm{diag}(d)), \quad \tilde{V} = V\exp(\hat{v}), \tag{1}$$

where $d \sim \mathcal{N}(0, 0.03^2 I)$ perturbs eigenvalues by a few percent and $v \sim \mathcal{N}(0, (3^\circ)^2 I)$ tilts the principal axes by a few degrees. The perturbed inertia is $\tilde{J} = \tilde{V}\tilde{D}\tilde{V}^T$.

The resulting perturbed principal moments are: $I_1 = 1.086 \times 10^8$, $I_2 = 1.275 \times 10^8$, $I_3 = 2.049 \times 10^8\,\mathrm{kg\,m^2}$. The angular momentum vector $J\boldsymbol{\omega}$ is misaligned from $\boldsymbol{\omega}$ by 1.75°, confirming $+Z$ is not a principal axis.

B. Desired Angular Velocity

The desired spin is $10\,\mathrm{RPM}$ about the solar panel normal ($+Z$ body):

$$\boldsymbol{\omega}_d = \frac{10 \times 2\pi}{60}\hat{z} = [0, 0, 1.0472]^T \ \mathrm{rad/s}. \tag{2}$$

C. Rotor Momentum via Superspin and Dynamic Balance

The rotor angular momentum $\boldsymbol{h}$ is computed to satisfy two conditions:

1) **Dynamic balance:** $\boldsymbol{\omega} \times (J\boldsymbol{\omega} + \boldsymbol{h}) = 0$, so the gyrostat equilibrium is torque-free.
2) **Superspin:** The effective spin-axis inertia λ satisfies $\lambda / I_{\perp,\max} \geq 1.2$, ensuring passive stability against transverse perturbations.

The effective spin-axis inertia is:

$$\lambda = \frac{(\boldsymbol{h} + J\boldsymbol{\omega}) \cdot \hat{\omega}}{|\boldsymbol{\omega}|} = 2.044 \times 10^8\,\mathrm{kg\,m^2}. \tag{3}$$

The maximum transverse inertia is $I_{\perp,\max} = 1.280 \times 10^8\,\mathrm{kg\,m^2}$, giving an inertia ratio of $\lambda / I_{\perp,\max} = 1.60 \geq 1.2$.

The computed rotor momentum is $\boldsymbol{h} = [+6.43 \times 10^6,\ +1.26 \times 10^6,\ 0]^T\ \mathrm{kgm^2/s}$ with $|\boldsymbol{h}| = 6.55 \times 10^6\ \mathrm{kgm^2/s}$. The dynamic balance residual $|\boldsymbol{\omega} \times (J\boldsymbol{\omega} + \boldsymbol{h})| < 10^{-10}$ confirms the equilibrium condition.

D. Gyrostat Simulation

The torque-free gyrostat equation is:

$$J\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (J\boldsymbol{\omega} + \boldsymbol{h}) = 0. \tag{4}$$

Coupled with quaternion kinematics, this is integrated using DOP853 (8th-order Dormand–Prince) with $\mathrm{rtol} = 10^{-12}$.

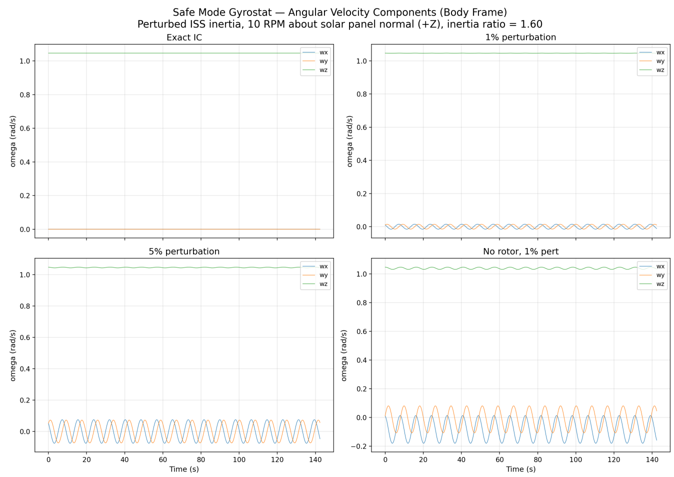

Fig. 1. Angular velocity components for four simulation cases. With the gyrostat rotor (Cases A–C), transverse rates remain bounded; without the rotor (Case D), the intermediate-axis instability causes divergence.

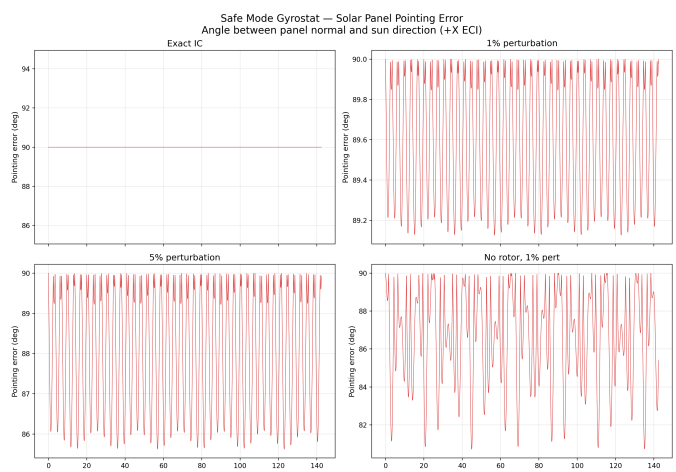

Fig. 2. Solar panel pointing error (angle between body $+Z$ and sun direction). The gyrostat maintains sub-degree pointing for Cases A–B and bounded error for Case C, while Case D (no rotor) diverges.

The estimated nutation period is

$$T_{\text{nut}} = \frac{2\pi}{\omega_s}\frac{\lambda}{|\lambda - I_\perp|} \approx 14.2\,\mathrm{s}. \tag{5}$$

Four cases are simulated for $10T_{\text{nut}} \approx 142\,\mathrm{s}$:

- **Case A:** Exact equilibrium initial conditions
- **Case B:** 1% transverse angular velocity perturbation
- **Case C:** 5% transverse perturbation
- **Case D:** No rotor ($\boldsymbol{h} = 0$), 1% perturbation

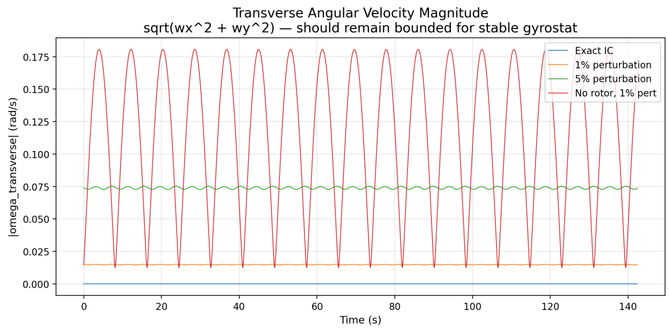

Fig. 3. Transverse angular velocity magnitude $\sqrt{\omega_x^2 + \omega_y^2}$. The gyrostat cases remain bounded (stable nutation) while Case D grows.

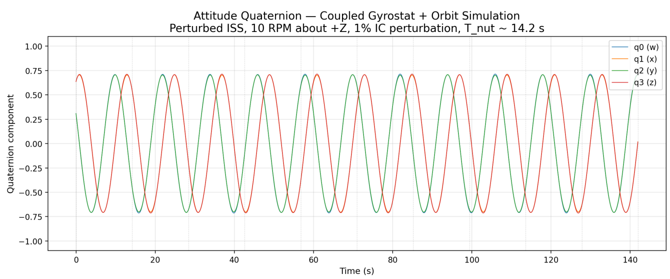

Fig. 4. Attitude quaternion components during the coupled gyrostat–orbit simulation over 10 nutation periods ($T_{\text{nut}} \approx 14.2\,\text{s}$).

Fig. 1 shows the angular velocity histories. Cases A–C demonstrate bounded nutation with the gyrostat rotor, while Case D (no rotor) shows the well-known intermediate-axis instability. Fig. 2 shows the corresponding pointing errors: the gyrostat maintains stable sun-pointing with bounded nutation even under 5% perturbation, whereas the uncontrolled case diverges rapidly.

II. Spacecraft Dynamics

Gyrostat dynamics are coupled to the orbit simulation from HW1. The state vector now includes position $\boldsymbol{R}$, velocity $\boldsymbol{V}$, attitude quaternion $\boldsymbol{q}$, angular velocity $\boldsymbol{\omega}$, and rotor momentum $\boldsymbol{h}$. The simulation uses the `mujoco_orbit` framework, which couples MuJoCo's multibody dynamics with orbital propagation including J_2, atmospheric drag, and solar radiation pressure.

Three reaction wheels (aligned with body X, Y, Z) store the rotor momentum computed in Section I. The same perturbed inertia and 1% transverse angular velocity perturbation are used.

The sun vector is assumed parallel to $+X$ in ECI coordinates. The initial attitude places the solar panel normal ($+Z$ body) toward the sun.

Fig. 4 shows the quaternion evolution, which exhibits the expected nutation oscillation. Fig. 5 shows the pointing error remains bounded below $\sim 10°$, consistent with the safe-mode analysis. The final pointing error is $8.89°$ with a maximum of $9.69°$, demonstrating that the gyrostat spin stabilization maintains approximate sun-pointing even in the presence of orbital coupling and environmental disturbances.

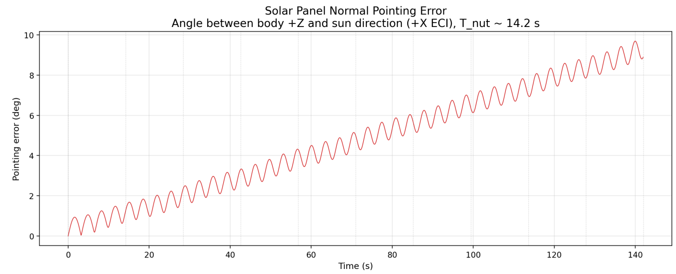

Fig. 5. Solar panel normal pointing error during coupled simulation. The error remains bounded below $\sim 10°$, consistent with the nutation amplitude from the 1% perturbation.

TABLE I
Attitude Sensor Specifications

Sensor	Parameter	Value
Gyroscope (GG1320AN)	ARW Bias instability	$0.002\,°/\sqrt{\text{hr}}$ $0.003\,°/\text{hr}$
Star Tracker (SED36)	Cross-boresight Roll	$5\,\text{arcsec}$ $25\,\text{arcsec}$
Sun Sensor (Adcole)	Accuracy	$0.05°$
Magnetometer (HMC2003)	Noise density At 10 Hz	$30\,\text{nT}/\sqrt{\text{Hz}}$ $94.9\,\text{nT}$
Horizon Sensor (Barnes 13-230)	Accuracy	$0.1°$

III. Attitude Sensors

Five attitude sensors are modeled based on ISS flight hardware and representative specifications from the literature.

A. Sensor Specifications

B. Covariance Derivations

Rate Gyroscope: The angle random walk (ARW) gives the white-noise spectral density: $\text{ARW} = 0.002\,°/\sqrt{\text{hr}} = 5.818 \times 10^{-7}\,\text{rad}/\sqrt{\text{s}}$. The per-sample rate noise standard deviation is $\sigma_\omega = \text{ARW}/\sqrt{\Delta t}$, yielding the covariance $R_{\text{gyro}} = (\text{ARW}^2/\Delta t)\, I_3$. Bias is modeled as a constant offset with $\sigma_b = 1.454 \times 10^{-8}\,\text{rad/s}$.

Star Tracker: Cross-boresight accuracy of $5\,\text{arcsec}$ ($\sigma_{xy} = 2.424 \times 10^{-5}\,\text{rad}$) and roll accuracy of $25\,\text{arcsec}$ ($\sigma_z = 1.212 \times 10^{-4}\,\text{rad}$). The angular error covariance in the sensor frame is $R_{\text{star}} = \text{diag}(\sigma_{xy}^2, \sigma_{xy}^2, \sigma_z^2)$, rotated to the body frame via the sensor mounting matrix.

Sun Sensor: Isotropic angular noise with $\sigma = 0.05° = 8.727 \times 10^{-4}\,\text{rad}$. The direction-vector covariance for Wahba's problem uses $\sigma_v^2 = \sigma^2$.

Magnetometer: At 10 Hz bandwidth, $\sigma_B = 30\sqrt{10} \approx 94.9\,\text{nT}$ per axis. The direction accuracy depends on field magnitude: $\sigma_{\text{dir}} = \sigma_B/|B|$. At $|B| = 35\,\mu\text{T}$: $\sigma_{\text{dir}} = 0.155°$.

Horizon Sensor: Isotropic angular noise with $\sigma = 0.1° = 1.745 \times 10^{-3}\,\text{rad}$.

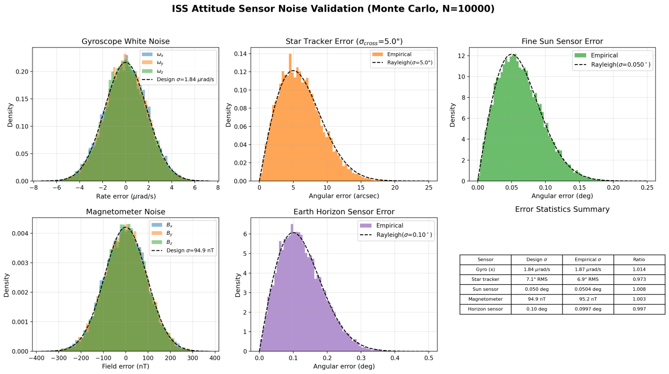

Fig. 6. Monte Carlo validation of sensor noise models. Histograms show empirical error distributions; dashed lines show the design distributions. All ratios (empirical/design σ) are within 1–2% of unity.

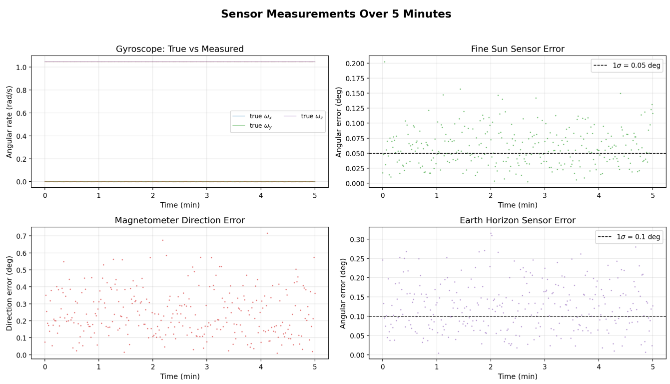

Fig. 7. Time series of sensor measurements over a 5-minute simulation. Gyro measurements track the true angular velocity with sub-μrad/s noise. Direction sensors show errors consistent with their design σ values.

C. Monte Carlo Validation

A Monte Carlo study with $N = 10{,}000$ samples validates that the simulated measurement errors match the design covariances.

Fig. 6 confirms all sensor noise models produce error statistics consistent with the design specifications. The empirical-to-design σ ratios are all within 2% of 1.0 across 10,000 trials.

IV. Static Attitude Estimation

Static attitude is estimated by solving Wahba's problem [2]:

$$\min_R \sum_i a_i \|\boldsymbol{b}_i - R\boldsymbol{r}_i\|^2, \quad a_i = 1/\sigma_i^2, \tag{6}$$

where $\boldsymbol{b}_i$ are body-frame observations and $\boldsymbol{r}_i$ are reference (ECI) directions.

A. Algorithms

Davenport's q-method [3]: The attitude profile matrix $B = \sum a_i \boldsymbol{b}_i \boldsymbol{r}_i^T$ is converted to the 4×4 Davenport matrix K. The optimal quaternion is the eigenvector of K corresponding to its maximum eigenvalue. Complexity: $O(1)$ after building B.

SDP relaxation [4]: Wahba's problem is reformulated as maximizing $\mathrm{tr}(KQ)$ subject to $Q \succeq 0$, $\mathrm{tr}(Q) = 1$. This semidefinite relaxation is tight for Wahba's problem (the optimal Q is always rank-1), so the SDP yields the exact solution. Solved using CLARABEL via CVXPY.

TABLE II
Wahba's Problem: Monte Carlo Results ($N = 5{,}000$)

Metric	q-method	SDP
Mean error (arcsec)	7.04	7.04
Median error (arcsec)	6.55	6.55
RMS error (arcsec)	7.81	7.81
Max error (arcsec)	23.80	23.80
Time per solve (ms)	0.024	1.613
Speedup	—	67× slower

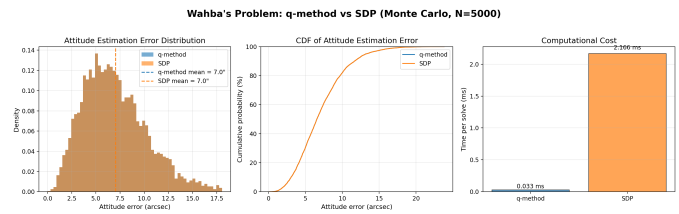

Fig. 8. Wahba's problem results. Left: error histograms (both algorithms are identical). Center: CDF of attitude error. Right: computational cost comparison showing the q-method is 67× faster.

B. Monte Carlo Comparison

Each trial uses five observations: two star directions, one sun direction, one nadir direction, and one magnetic field direction. The observation weights are $1/\sigma_i^2$ based on each sensor's angular noise.

Table II and Fig. 8 show that both algorithms produce identical attitude estimates (mean error $7.04\,\mathrm{arcsec}$, RMS $7.81\,\mathrm{arcsec}$). This confirms the SDP relaxation is tight, as expected for Wahba's problem. The q-method is 67× faster ($0.024\,\mathrm{ms}$ vs. $1.613\,\mathrm{ms}$ per solve), making it the preferred choice for real-time applications.

V. Changelog

Date	Description
2026-03-12	HW2: Safe mode, dynamics, sensors, estimation
2026-02-26	HW1: Mission description, model, dynamics, Euler

References

[1] NASA, "International Space Station: Facts and Figures," https://www.nasa.gov/feature/facts-and-figures, 2024.
[2] G. Wahba, "A least squares estimate of satellite attitude," *SIAM Review*, vol. 7, no. 3, p. 409, 1965.
[3] P. B. Davenport, "A vector approach to the algebra of rotations with applications," NASA TN D-4696, 1968.
[4] F. L. Markley and J. L. Crassidis, *Fundamentals of Spacecraft Attitude Determination and Control*. New York: Springer, 2014, ch. 5.
[5] Honeywell, "GG1320AN Digital Ring Laser Gyroscope," datasheet, 2020.
[6] Sodern, "SED36 Star Tracker," product specification, 2019.
[7] Adcole Corporation, "Two-Axis Digital Sun Sensor," datasheet.
[8] Honeywell, "HMC2003 Three-Axis Magnetic Sensor Hybrid," datasheet, 2018.
[9] J. R. Wertz and W. J. Larson, *Space Mission Analysis and Design*, 3rd ed. Microcosm Press, 1999.